

Unit 10, Lesson 2: Identifying Domain and Range

Benchmark

MA.8.F.1.1 Given a set of ordered pairs, a table, a graph, or mapping diagram, determine whether the relationship is a function. Identify the domain and range of the relation.

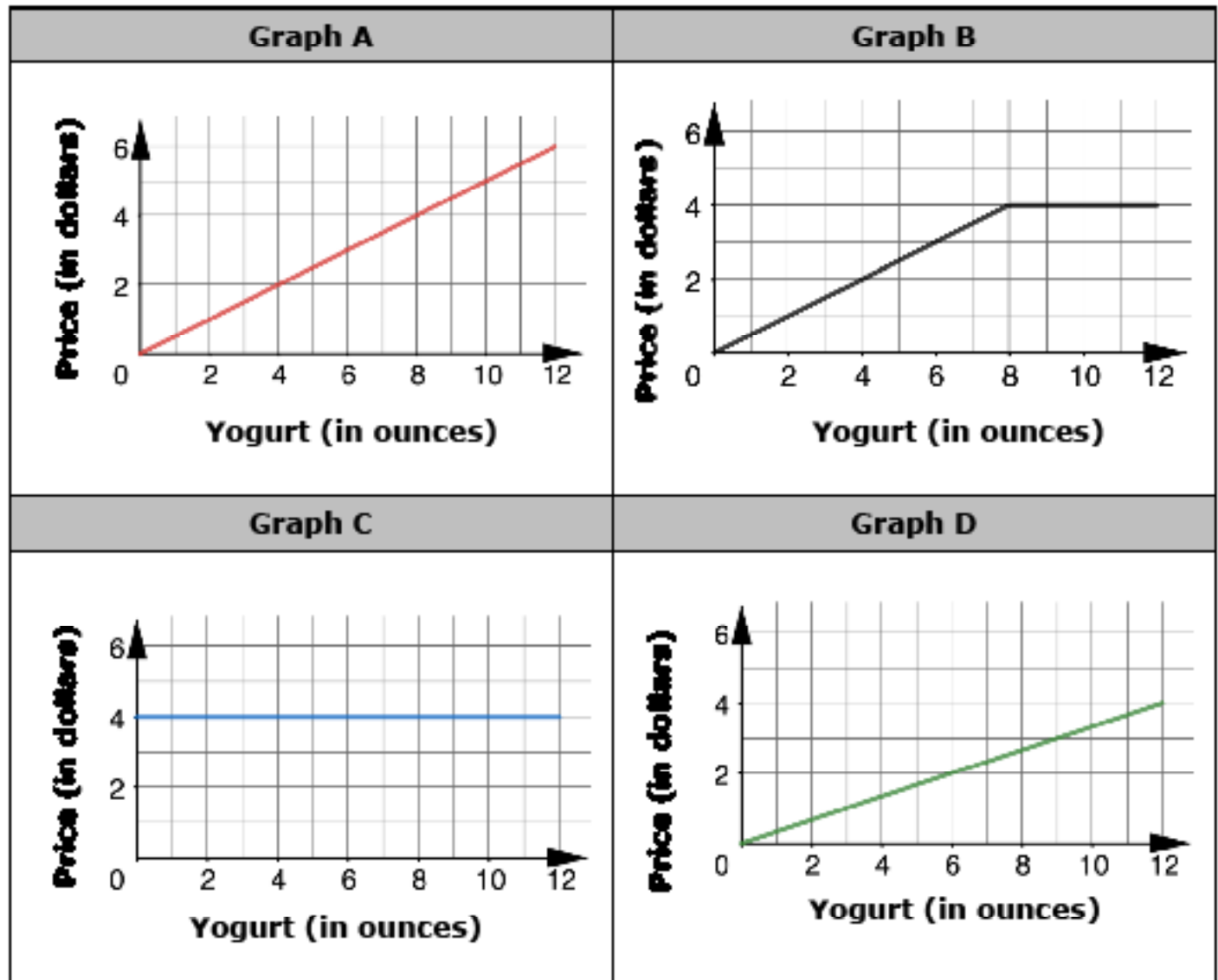
Learning Targets

- ☐ I can relate domain and range to independent and dependent variables in a relation.
- ☐ I can identify the domain and range of a relation.

10.2.1 Warm-Up: Math Talk

A self-serve frozen yogurt store sells servings up to 12 ounces. It charges \$0.50 per ounce for a serving between 0 and 8 ounces, and \$4.00 for any serving greater than 8 ounces and up to 12 ounces.

Four graphs are shown.



1. Explain which graph best represents the price as a function of the weight of a serving of yogurt.

10.2.2 Guided Instruction: Identifying the Domain and Range of a Relation

1. Consider the statements shown.

$$12 \div 3 = 4 \text{ because } 12 = 4 \cdot 3.$$

$$6 \div 0 = x \text{ because } 6 = x \cdot 0.$$

In the second statement, what value can be used in place of x so that the statement is true?

The inputs of a relation are described as its **domain**.



The **domain** is the complete set of possible values of the input of a function or relation. The domain may vary depending on the context.

Sometimes one or more values are not allowed in the domain.

2. Use the rule to complete the table of values to represent the relation described by the rule and answer the subsequent questions.

a. Complete the table of values.

Rule A	Table of Values												
$x \Rightarrow$ divide 1 by the input $\Rightarrow \frac{1}{x}$	<table><tr><th>Input</th><th>Output</th></tr><tr><td>$-\frac{2}{7}$</td><td>$-\frac{7}{2}$</td></tr><tr><td>1</td><td></td></tr><tr><td>$\frac{4}{5}$</td><td></td></tr><tr><td>0</td><td></td></tr><tr><td>12</td><td></td></tr></table>	Input	Output	$-\frac{2}{7}$	$-\frac{7}{2}$	1		$\frac{4}{5}$		0		12	
Input	Output												
$-\frac{2}{7}$	$-\frac{7}{2}$												
1													
$\frac{4}{5}$													
0													
12													

b. In this relation, _____ is not an allowable input.

Therefore, the domain does not include _____.

c. Work with your partner to determine if there is another input that is not allowable for $\frac{1}{x}$.

The domain of Rule A can be represented using inequalities for the input values defined as x .

$x > \underline{\hspace{1cm}}$ or $x < \underline{\hspace{1cm}}$

These two inequalities can also be written as one, $x \neq \underline{\hspace{1cm}}$.

The outputs of a relation are described as its *range*.



The **range** is the complete set of possible values of the output of a relation or function.

3. Use the rule to complete the mapping diagram and answer the subsequent questions.
- a. Complete the mapping diagram.

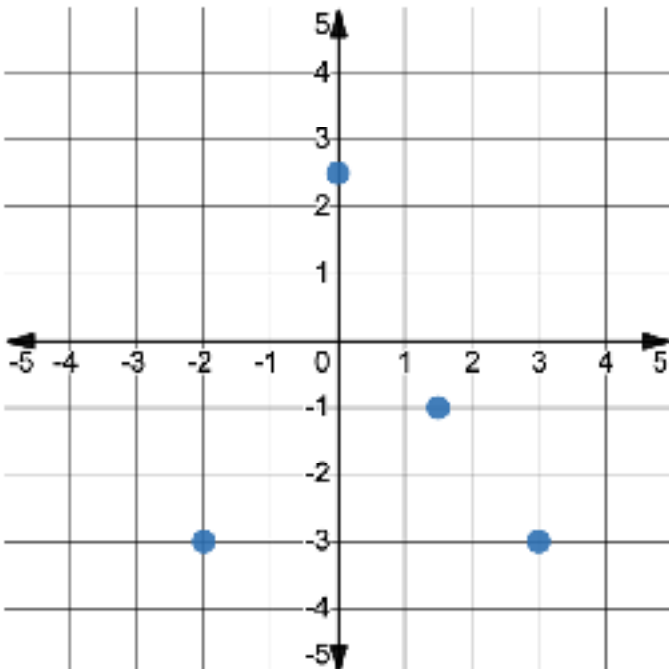
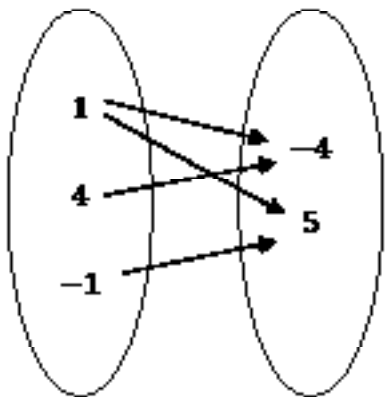
Rule B	Mapping Diagram
$x \rightarrow$ square the input $\rightarrow y$	x -4 0 9 y

- b. Based on the rule, what type of output values are unattainable for this relation?
- c. Write an inequality to represent the range of this relation.

$y \bigcirc \underline{\hspace{1cm}}$

At times, the domain and range of a relation can be represented using a list of numbers.

4. For each relation, write the domain and range as a set of numbers.

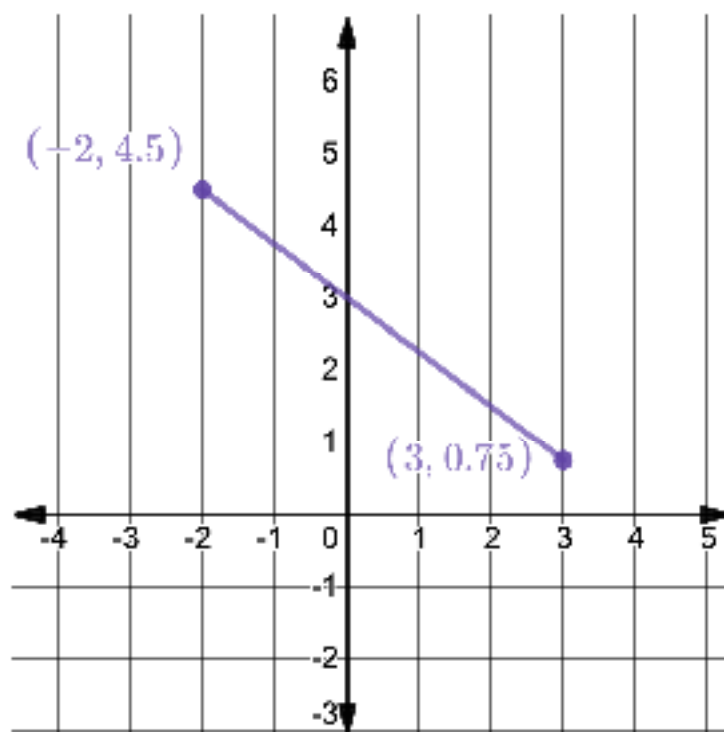
Relation A	Relation B
 Domain: _____ Range: _____	 Domain: _____ Range: _____

There are also instances where the domain and range of a relation can be represented by a compound inequality.



A **compound inequality** is a conjunction of two or more inequalities.

For example, $-4 \leq x \leq \frac{3}{5}$ is a compound inequality that is read, “ -4 is less than or equal to x , which is less than or equal to $\frac{3}{5}$.” This inequality is true for any value of x between -4 and $\frac{3}{5}$, including either of those values.

5. A graph is shown.

- a. Represent the domain and range of the graph using compound inequalities.

Domain: _____

Range: _____

- b. Compare this graph to Relation A in the previous problem. Discuss with your partner why representing the domain and range of this graph could not be done by writing the elements of each in a list.

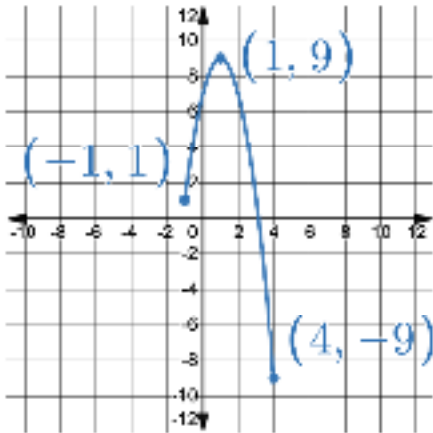


Representing Domain and Range

When the elements of the domain or range consist of a **set number of values**, each can be **represented by a list**. When the elements of the domain or range consist of an **infinite number of points between specified values**, an **inequality** is best.

10.2.3 Try It: Identifying the Domain and Range of a Relation

1. Identify the domain and range of each relation. Represent the domain as a list or using an inequality.

Relation	Domain										
<table><tr><th>x</th><th>y</th></tr><tr><td>-4.7</td><td>7</td></tr><tr><td>2.4</td><td>4.7</td></tr><tr><td>-1.6</td><td>3.9</td></tr><tr><td>8</td><td>10</td></tr></table>	x	y	-4.7	7	2.4	4.7	-1.6	3.9	8	10	Range
x	y										
-4.7	7										
2.4	4.7										
-1.6	3.9										
8	10										
Relation	Domain										
$\{(2.3, 7.4), (3.8, 6), (1.9, 5.2), (2.3, 8)\}$	Range										
Relation	Domain										
$x \Rightarrow \boxed{\text{divide 5 by the input}} \Rightarrow \frac{5}{x}$	Range										
Relation	Domain										
	Range										

Complete each statement.

1. Today I am still unsure about...
2. To fill this gap, I intend to...

